

Synent Technologies - Data Science Internship

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Task 4: CSV to Dashboard Report

1. Objective

The objective of Task 4 was to build an interactive dashboard capable of ingesting raw CSV datasets and dynamically generating exploratory data analysis (EDA) charts. This fulfills the requirement of creating a dynamic visualization tool for rapid dataset profiling.

2. Application Features

1. Dynamic File Upload: Users can upload any CSV file using the Streamlit file uploader.
2. Dataset Overview: The application automatically parses column data types, calculates missing values, and generates comprehensive descriptive statistics.
3. Visual Exploration Studio: A smart plotting engine built with Plotly. It automatically distinguishes between numeric and categorical features, allowing the user to seamlessly generate Scatter Plots, Box Plots, Bar Charts, Histograms, and Correlation Heatmaps.
4. Sample Datasets: Built-in support for the Titanic and Iris datasets to allow instant testing and verification.

3. Technical Stack

- Frontend & App Logic: Streamlit
- Data Manipulation: Pandas
- Visualization: Plotly Express
- Language: Python 3.x

4. Evaluation & Compliance

The application perfectly adheres to the Synent guidelines for Task 4. It operates reliably, prevents user errors (such as attempting to plot histograms on categorical text columns), and maintains a highly professional, dark-themed UI. The code is fully documented and deployed to a public GitHub repository.